

EXAM 7 – SPRING 2011

7. (1.25 points)

Given the following cumulative reported loss emergence pattern:

<u>Accident Year Age in Months</u>			
<u>12</u>	<u>24</u>	<u>36</u>	<u>48</u>
15%	35%	65%	80%

- The loss emergence follows an over-dispersed Poisson model, with dispersion parameter, ϕ , equal to 1.3.
- Expected ultimate losses for accident year 2009 are \$150,000,000.

a. (0.75 point)

Calculate the variance of the 24 to 36 month incremental reported losses for accident year 2009.

b. (0.5 point)

Explain why modeling paid losses or claim counts may be preferable to modeling reported losses when using an over-dispersed Poisson model.

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Sample Solutions
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Solution 1

a)

Expected Ultimate Loss = 150,000,000

2009 24-36 development = .65 - .35 = .3

Expected 24-36 development = 150,000,000 $\times$.3 = 45,000,000

Over Dispersed Poisson Variance = Dispersion factor $\times$ Mean = 1.3 $\times$ 45,000,000
=58,500,000

b)

Paid losses and claim counts are not as susceptible to having negative incremental development. Paid losses and claim counts are not affected by changes in the case reserve estimates that affect incurred losses. A limitation of the over dispersed Poisson distribution is the sum of incremental claims in column j must also be positive.